1 Lit review

1. **Foundations of Belief ~~asd~~and Updating & Variational Inference**
   - A general framework for updating belief distributions (Bissiri et al.)
   - Stochastic Variational Inference
   - An optimisation-centric view on Bayes' rule: Reviewing and generalising variational inference (Knoblauch et al.)
2. **The Shift to Predictively Oriented Posteriors**
   - Predictive Variational Inference: Learning the predictively optimal posterior distribution
   - Predictive Variational Inference for flexible regression models
   - Predictively Oriented Posteriors (McLatchie et al.)
3. **Particle-Based Inference, MMD, and Misspecification**
   - Stein Variational Gradient Descent (Liu & Wang) (Very quick mention) and time complexity things
   - A Rigorous Link between Deep Ensembles and (Variational) Bayesian Methods
   - Prediction-Centric Uncertainty Quantification via MMD (Shen et al.), Wild et al. 2023
   - maybe: (Detecting model misspecification in Bayesian inverse problems via variational gradient descent)
4. **Variational Inference with Normalising Flows?? Adaptive Heterogeneous Mixtures of Normalising Flows for Robust Variational Inference?? And more MOE papers...**

2 1:

~~In this section we will begin by outlining some of the common notation with further notation introduced in the literature review. . . .~~

At the core of statistical inference is the challenge of mathematically capturing the structure and uncertainty inherent in observed data. . . .

Suppose that we have a statistical model $P_\theta \in \mathscr{P}$, where $\mathscr{P} = \{P_\theta : \theta \in \Theta\}$, with the parameters $\theta \in \Theta$ for the data points $x_{1:n} = \{x_1, x_2 \ldots, x_n\}$ with a theoretical and often unknown data generating process (DGP) $D(x)$. The parameters θ index the candidate distributions ~~used to describe the data~~, with their specific values typically encoding information about the data itself. In frequentist statistics these parameters θ are treated as fixed but unknown constants, however, when dealing with the Bayesian approach, the parameters are treated as random variables / distributions. In the most standard Bayes we have the posterior density

$$\pi_B(\theta \mid x_{1:n}) = \frac{\pi(\theta) L(x_{1:n} \mid \theta)}{m(x_{1:n})}.$$

Where $\pi(\theta)$ is our prior knowledge of the parameters θ, $L(x_{1:n} \mid \theta)$ is the likelihood function representing the probability of the observed data under those parameters, and $\pi_B(\theta \mid x_{1:n})$ is the pdf of the distribution $\theta \mid x_{1:n}$. The B here indicating that this is the standard Bayesian posterior. Here $m(x_{1:n}) = \int_\Theta L(x_{1:n} \mid \theta) d\pi(\theta)$ is the marginal normalising constant of the distribution, whose computation often makes the distribution intractable.

Bayesian inference is conceptually and practically appealing as, unlike frequentist statistical methods, we are able to allow our previous knowledge, specific domain expertise and experts in the subject matter to further inform our distribution. This injection of previous knowledge comes in the form of a carefully specified prior $\pi(\theta)$. Naturally, this comes with the potential risk of poor prior choice.

...

Further, Bayesian posteriors produce belief distributions (as opposed to point estimates, as an example, the direct outputs of MLE) over the parameters of interest $\theta \in \Theta$.

This carries two related benefits. First, prediction propagates parameter uncertainty rather than conditioning on a single value of θ; where the conventional Bayesian predictive density for an unobserved x_{new} integrates the model over the posterior,

$$p(x_{\text{new}} \mid x_{1:n}) = \int_{\Theta} p(x_{\text{new}} \mid \theta)\pi_B(\theta \mid x_{1:n})d\theta. \tag{1}$$

Second, any quantity of interest is recovered as a functional of the posterior, for instance the posterior mean $\hat{\theta}$ is recovered as $\int_{\Theta} \theta \pi_B(\theta \mid x_{1:n})d\theta$. Because inference averages over θ instead of fixing it at a single best[1] value, the resulting predictions are less prone to the over-confidence of a plug-in point estimate.

Moving to a more generalised Bayes, we can ~~change this to~~recast inference as an optimisation-centric approach. Generalised Bayes replaces the likelihood with a loss function raised in an exponential.

$$\begin{aligned}
\pi_w(\theta \mid x_{1:n}) &= \frac{\exp\left\{-w\sum_{i=1}^{n}\ell(\theta,x_i)\right\}\pi(\theta)}{\displaystyle\int_{\Theta}\exp\left\{-w\sum_{i=1}^{n}\ell(\theta,x_i)\right\}\pi(\theta)d\theta} \\
&\propto \exp\left\{-w\sum_{i=1}^{n}\ell(\theta,x_i)\right\}\pi(\theta)
\end{aligned}$$

where ~~once again there is a normaliser in the denominator that without we can find the distribution up to proportionality~~, as before, the denominator is the normalising constant; without it we characterise the distribution only up to proportionality. There also exists a learning rate / loss scale $w \in \mathbb{R}_{>0}$ (strictly positive; w rescales the total loss). This posterior is the Bissiri–Holmes–Walker (2016) Gibbs posterior.

It is important to note here that from this posterior we can recover our previous standard Bayes posterior π_B in the specific case with $\ell(\theta,x_i) = -\log p(x_i \mid \theta)$ and the learning rate $w = 1$. The loss $-\log p(x_i \mid \theta)$ is the self-information (surprisal) of each observation, so this choice penalises the model exactly by how surprised it is by the data. Following this we get

$$\begin{aligned}
\pi_w(\theta \mid x_{1:n}) &\propto \exp\left\{-w\sum_{i=1}^{n}\ell(\theta,x_i)\right\}\pi(\theta) \\
&= \exp\left\{\sum_{i=1}^{n}\log p(x_i \mid \theta)\right\}\pi(\theta) \\
&= \left(\prod_{i=1}^{n}p(x_i \mid \theta)\right)\pi(\theta) \\
&= L(x_{1:n} \mid \theta)\pi(\theta)
\end{aligned}$$

And further the normalising constant recovers the marginal likelihood in the same manner. Hence we recover $\pi_B(\theta \mid x_{1:n})$. So standard Bayes is not a separate paradigm but a single specific instance of a more generalised approach where $\ell = \log$ loss, and $w = 1$.

[1] Here *best* refers to the best available point estimate of θ, e.g. the MLE or MAP.

Aside: the learning rate w

Looking at the learning rate w here, we can see this as a weighting between the prior and the data, and a guard against model misspecification (assuming the prior is well specified). It acts as an effective-sample-size multiplier, effectively updating the prior with only wn data points. The limiting behaviour makes this concrete:

$$w \to 0^+ : \pi_w \to \pi \text{ (prior)}; \qquad w = 1, \text{ log loss} : \pi_w = \pi_B; \qquad w \to \infty : \pi_w \to \delta_{\hat{\theta}}, \ \hat{\theta} = \arg\min_\theta \sum_i \ell(\theta, x_i).$$

The effect of this learning rate is exact and analytic and *does* affect the final distribution of the posterior. Here w is part of the target distribution itself, not of any algorithm used to approximate it. This should be held separate from the step size of an optimiser, which (for a non-analytic solver) governs the trajectory toward a fixed target rather than the target. This distinction is flagged now because as later this property is altered. Once the posterior is itself defined as the solution of a particle optimisation (PVI), the data-weight and the optimiser's effect on the realised distribution no longer cleanly separate. Further discussed in Section **??**.

Coherence and the loss-based view

Bissiri, Holmes & Walker do not posit the exponentiated-loss form; they derive it from a *coherence* requirement on how beliefs may be updated. Coherence demands that the update be self-consistent under partitioning of the data: updating on a first batch and then on a second must yield the same belief as updating on the pooled sample at once, so that the order and grouping of the data cannot change our conclusions. An update with (x_1, x_2) together or $\{(x_1), (x_2)\}$ separatatly yeilds the same result. This coherence property can be formally expressed as

$$\psi[l(\theta, x_2), \psi\{l(\theta, x_1), \pi(x)\}] = \psi\{l(\theta, x_2), +l(\theta, x_1), \pi(x)\},$$

where the function ψ is the update function such that in the standard case

$$\pi(\theta \mid x) \equiv \psi\{l(\theta; x), \pi(\theta)\}.$$

For an additive loss they show the unique update satisfying this is the Gibbs posterior above. This matters going forward because coherence is a property of the loss: the self-information (log) loss satisfies it, but as we adopt richer, predictive and discrepancy-based losses later, coherence is no longer automatic and must be checked rather than assumed.

Some of the motivation for the loss-based view is that one may often only be interested in a simple, low-dimensional statistic, such as a population mean or median. In standard Bayesian frameworks, estimating these simple statistics typically (As in pi_B) requires modelling the entire data distribution. Furthermore, some parameters of interest do not directly index any standard parametric family of density functions. (update / check mroe LATER) A loss can connect a parameter to data directly In our setting later on, we nonetheless retain the full posterior rather than a single functional, as ...

[2] [3]

The optimisation-centric view and the Rule of Three

The generalised bayes posterior π_w has an equivalent expression focusing more on the optimisation side. In Knoblauch, Jewson & Damoulas (2022) it is shown that ~~and setting the multiplicative updating rule aside,~~ the exact posterior can equally be obtained as the solution of an infinite-dimensional optimisation problem over the space $\mathscr{P}(\Theta)$ of *all* probability measures on Θ.

[2] absolute-error-loss inference can still succumb to misspecification, a robust loss limits but does not remove sensitivity to the DGP

[3] Link to PrOs, also the problem of w

~~refrences? maybe as called out : representation they trace to Csiszár (1975), Donsker & Varadhan (1975) and Zellner (1988):~~

$$\pi_B(\theta \mid x_{1:n}) = \arg\min_{q\in\mathscr{P}(\Theta)} \left\{ \mathbb{E}_q\Big[\sum_{i=1}^{n}\ell(\theta,x_i)\Big] + \mathrm{KL}(q\|\pi)\right\}, \qquad \ell(\theta,x_i) = -\log p(x_i \mid \theta).$$

Replacing the log loss with a general ℓ (and absorbing the loss scale w into ℓ) returns exactly the generalised Gibbs posterior of the previous section; Bissiri et al. (2016) show the minimiser exists and coincides with π_w[4]whenever the normaliser $Z = \int_\Theta \exp\{-\sum_i \ell(\theta,x_i)\}\pi(\theta)d\theta$ is finite. The objective can be interpreted as regularised empirical risk minimisation; it functions in two parts, minimising the expected loss over the data while the KL term pulls the solution toward the prior. These are both important points that will be further discussed later on. It should be noted that here it is the KL term which causes the optimisation problem to converge to a distribution, without it the problem collapses to a point mass $\delta_{\hat\theta}$ at the empirical risk minimiser $\hat\theta = \arg\min_\theta \sum_i \ell(\theta,x_i)$. This can be seen as we derive our previous bayes posteriors from this optimisation results.

Letting $\ell_n(\theta) = \sum_{i=1}^{n}\ell(\theta,x_i)$ denote the cumulative loss, the normalising constant Z and the normalised Gibbs posterior $\pi_w(\theta)$ are defined as

$$Z = \int_\Theta \pi(\theta)e^{-\ell_n(\theta)}d\theta \quad \text{and} \quad \pi_w(\theta) = \frac{\pi(\theta)e^{-\ell_n(\theta)}}{Z}.$$

The objective then collapses into a single Kullback-Leibler (KL) divergence,

$$\begin{aligned}
\mathbb{E}_q\left[\sum_{i=1}^{n}\ell(\theta,x_i)\right] + \mathrm{KL}(q\|\pi) &= \int_\Theta q(\theta)\ell_n(\theta)d\theta + \int_\Theta q(\theta)\log\frac{q(\theta)}{\pi(\theta)}d\theta \\
&= \int_\Theta q(\theta)\left[\log\frac{q(\theta)}{\pi(\theta)} + \ell_n(\theta)\right]d\theta \\
&= \int_\Theta q(\theta)\log\frac{q(\theta)}{\pi(\theta)e^{-\ell_n(\theta)}}d\theta \\
&= \int_\Theta q(\theta)\log\frac{q(\theta)}{Z\pi_w(\theta)}d\theta \\
&= \mathrm{KL}(q\|\pi_w) - \log Z.
\end{aligned}$$

Since $\log Z$ is independent of q and $\mathrm{KL}(q\|\pi_w) \geq 0$ with equality if and only if $q = \pi_w$, the minimiser over $\mathscr{P}(\Theta)$ is exactly the Gibbs posterior,

$$\arg\min_{q\in\mathscr{P}(\Theta)} \left\{ \mathbb{E}_q\left[\sum_{i=1}^{n}\ell(\theta,x_i)\right] + \mathrm{KL}(q\|\pi)\right\} = \pi_w(\theta) = \frac{\pi(\theta)\exp\{-\sum_{i=1}^{n}\ell(\theta,x_i)\}}{Z}.$$

Specialising to the log loss $\ell(\theta,x_i) = -\log p(x_i \mid \theta)$ yields $e^{-\ell_n(\theta)} = \prod_{i=1}^{n} p(x_i \mid \theta) = L(x_{1:n} \mid \theta)$ alongside the marginal likelihood $Z = \int_\Theta \pi(\theta)L(x_{1:n} \mid \theta)d\theta = m(x_{1:n})$. Substituting these components gives

$$\pi_w(\theta) = \frac{\pi(\theta)L(x_{1:n} \mid \theta)}{m(x_{1:n})} = \pi_B(\theta \mid x_{1:n}),$$

which recovers the standard Bayesian posterior. A quick proof of dericlt mass for without KL?

Furthermore, we can see now that by using the Bayesian posterior, we are equivalently committing to this specific optimisation problem. As [paper ...] points out, similar to the generalised Bayes, using this standard posterior is only the right tool if minimising this objective actually reflects the overarching goal of the inference task. If the standard log loss does not align with those practical goals, this framework justifies replacing it with a custom loss function. (Knoblauch, Jewson & Damoulas, 2022).

[4]Knoblauch et al. (2022) do not notationally distinguish the standard and generalised Bayesian posteriors, writing q_B^* for both; here we keep them separate as π_B and π_w.

This transformation to an optimisation problem is what we will focus on thoughout this thesis, and from here on out.

Taking inference as an optimisation, the standard posterior is seen to fix three choices that need not be fixed. A loss ℓ, the divergence KL measuring departure from the prior, and the feasible set $\mathscr{P}(\Theta)$ optimised over. Knoblauch et al. free all three, defining the *Rule of Three* (RoT) posterior

$$\pi^{\star}_{(\ell,D,\Pi)} := \arg\min_{q\in\Pi}\left\{\mathbb{E}_q\Big[\sum_{i=1}^{n}\ell(\theta,x_i)\Big]+D(q\|\pi)\right\},$$

specified by a loss ℓ, a divergence D, and a feasible set $\Pi \subseteq \mathscr{P}(\Theta)$[5]. Standard Bayes is the choice $\pi^{\star}_{(-\log p,\mathrm{KL},\mathscr{P}(\Theta))}$ and generalised Bayes is $\pi^{\star}_{(\ell,\mathrm{KL},\mathscr{P}(\Theta))}$; the same template also recovers tempered and PAC-Bayesian posteriors. The three arguments ~~are not arbitrary: they~~ answer three assumptions that justify standard Bayes, **but may have issues** . A well-specified prior (P), a well-specified likelihood (L), and unlimited computation (C). The divergence D controls how, and how strongly, the prior is enforced (P); the loss ℓ controls robustness to model misspecification (L); and the feasible set Π controls tractability (C).

An issue arieses in this setup where, computationally, this optimisation becomes quite intractable over an infinite dimensional space; the third argument in our objective Π as he space of all possible probability measures $\mathscr{P}(\Theta)$, is infinite dimensional. Variational Inference helps this by restricting this third argument to a finite-dimensional family $\mathscr{Q} = \{q(\theta \mid \kappa) : \kappa \in K\}$, VI shrinks an impossible infinite-dimensional search down to a tractable finite one. Thus while VI is an approximation of the true posterior, it is still shown to be grounded by the same optimisation problem but constrained, yielding

$$\pi^{\star}_{\mathrm{VI}} = \pi^{\star}_{(\ell,\mathrm{KL},\mathscr{Q})}.$$ [6]

Standard VI is then not a separate approximation principle but the *best $\mathscr{Q}$-constrained solution of the very same objective* (Knoblauch et al., 2022, Thm. 2). The tractable case in which $\Pi = \mathscr{Q}$ is a variational family and D may differ from KL is what they call *Generalised Variational Inference* (GVI). This shifts the role of $\mathscr{Q}$; rather than treating inference in $\mathscr{Q}$ as an approximation to π_B, we may take the objective itself as the design target and construct belief distributions in $\mathscr{Q}$ with the properties we want from the outset. MoE-PVI is in this frame as a choice of feasible set Π (mixtures of experts) together with a [predictive / discrepancy-based] loss ℓ ...

Variational and stochastic variational inference

Whether working within a standard or generalised framework, the resulting posterior typically remains intractable due to the normalising constant $p(x_{1:n})$, often referred to as the marginal likelihood or model evidence. This evidence represents the total probability of observing the data marginalised over all possible parameter configurations, calculated as $p(x_{1:n}) = \int_{\Theta} p(x_{1:n} \mid \theta)\pi(\theta)d\theta$. Because this high-dimensional integral rarely admits a closed-form solution, directly computing the posterior is usually impossible. Variational inference attempts to sidestep this integration bottleneck by restricting the search to a tractable family of distributions $\mathscr{Q}$ and selecting the member that is closest to the true posterior in Kullback-Leibler divergence. The ideal variational distribution is therefore

$$q^{\star} = \arg\min_{q\in\mathscr{Q}} \mathrm{KL}(q(\theta)\|\pi_B(\theta \mid x_{1:n})).$$

However, this divergence cannot be minimised directly because its evaluation depends on the very intractable evidence we are trying to avoid. Expanding the definition of the KL divergence reveals

[5] Notation: Knoblauch et al. write the RoT posterior as $P(\ell,D,\Pi)$. Here it is $\pi^{\star}_{(\ell,D,\Pi)}$

[6] Knoblauch et al., 2022, Thm. 2

this dependency explicitly, yielding

$$\begin{aligned}\mathrm{KL}(q(\theta)\|\pi_B(\theta\mid x_{1:n})) &= \int_\Theta q(\theta)\log\frac{q(\theta)}{\pi_B(\theta\mid x_{1:n})}d\theta \\ &= \int_\Theta q(\theta)\log\frac{q(\theta)p(x_{1:n})}{p(x_{1:n}\mid\theta)\pi(\theta)}d\theta \\ &= \int_\Theta q(\theta)\left[\log q(\theta)-\log\pi(\theta)-\log p(x_{1:n}\mid\theta)\right]d\theta+\log p(x_{1:n}) \\ &= \mathrm{KL}(q(\theta)\|\pi(\theta))-\mathbb{E}_q[\log p(x_{1:n}\mid\theta)]+\log p(x_{1:n}).\end{aligned}$$

By rearranging these terms, we can express the log evidence as the sum of two distinct components,

$$\log p(x_{1:n}) = (\mathbb{E}_q[\log p(x_{1:n}\mid\theta)]-\mathrm{KL}(q(\theta)\|\pi(\theta)))+\mathrm{KL}(q(\theta)\|\pi_B(\theta\mid x_{1:n})).$$

The bracketed term is known as the Evidence Lower Bound (ELBO). Because the KL divergence is strictly non-negative, the ELBO strictly lower-bounds the true log evidence. Furthermore, since the log evidence $p(x_{1:n})$ is a constant with respect to the variational distribution q, minimising the KL divergence to the exact posterior is mathematically identical to maximising this lower bound, formally defined as

$$\mathrm{ELBO}(q) = \mathbb{E}_q[\log p(x_{1:n}\mid\theta)]-\mathrm{KL}(q(\theta)\|\pi(\theta)).$$

Classical coordinate-ascent variational inference optimises this ELBO with a full sweep over the dataset at each iteration, which quickly becomes computationally prohibitive for large datasets. To address this, Stochastic Variational Inference (Hoffman et al., 2013) applies stochastic optimisation by forming noisy, unbiased natural-gradient estimates of the ELBO from random data minibatches. By updating the parameters under a Robbins–Monro step-size schedule requiring $\sum_t \rho_t = \infty$ and $\sum_t \rho_t^2 < \infty$, the algorithm scales efficiently to massive datasets. While this natural gradient takes a remarkably simple form for conditionally conjugate exponential-family models, black-box variants (Ranganath et al., 2014) remove this conjugacy requirement entirely by relying on score-function gradient estimators. However, the defining limitation for all of these approaches is that they fix a parametric family $\mathscr{Q}$ in advance. The quality of the final approximation can never exceed the expressiveness of this chosen family, which is precisely the rigid restriction that the particle-based methods discussed in Section 3 are designed to relax.

3 2:

McLatchie et al. (2025) defining the predictively oriented (PrO) posterior as

Rigorous Link between Deep Ensembles and (Variational) Bayesian Methods New statistical principle that combines desirable aspects of both parameter inference and density estimation. Uncertainty as a consequence of predictive ability. The discrepancy from (1).

PrO posteriors converge at $n^{-\frac{1}{2}}$ while conventional only converge at this rate if the model can recover the true DGP.

$$\pi_w = \arg\min_{q\in\mathscr{P}(\Theta)}\left\{\frac{\lambda_n}{n}\sum_{i=1}^n\int_\Theta S(P_\theta,x_i)dq(\theta)\;+\;\mathrm{KL}(q\|\pi)\right\}$$

$$\pi_{\mathrm{PrO}} = \arg\min_{q\in\mathscr{P}(\Theta)}\left\{\frac{\lambda_n}{n}\sum_{i=1}^n S(P_q,x_i)\;+\;\mathrm{KL}(q\|\pi)\right\},\qquad P_q := \int_\Theta P_\theta dq(\theta) \tag{2}$$

$$d\pi_{\mathrm{PrO}}(\theta)\;\propto\;\pi(\theta)\exp\left\{-\frac{\lambda_n}{n}\sum_{i=1}^n\left.\frac{\delta S(P_q,x_i)}{\delta q}\right|_{q=\pi_{\mathrm{PrO}}}(\theta)\right\} \tag{3}$$

Gibbs minimises the average of scores; PrO minimises the score of the average.

4 3:

At the end of Section 1 we saw a structural limitation with standard Variational and stochastic variational inference. That is, to make our approximation tractable we fix a parametric family $\mathscr{Q}$ in advance. However, by setting an approximating distribution we limit how rich our expression can be to the maximum the selected family permits; under model misspecification, where the best description of the data may be multimodal or heavy-tailed, this restriction is what degrades the approximation.

With particle-based methods (or particle approximations), we remove this constraint. Instead of parameterising q (where q is our variational approximation of the true target distribution π_w as noted in section 1 (elbo)) and optimising q's parameters, we use a cloud of M particles $\{\theta_1, \ldots, \theta_M\}$ with the empirical measure $q \approx \frac{1}{M}\sum_j \delta_{\theta_j}$ and transport the particles directly. By transitioning to particle-based methods, we redefine the feasible set Π within the Rule of Three framework, replacing the rigid parametric $\mathscr{Q}$ with the infinitely flexible space of empirical measures.

Two complementary lineages dominate the literature: the deterministic Stein flow of Liu and Wang (2016), and the Wasserstein-gradient-flow view of Wild et al. (2023), which unifies deep ensembles, Langevin dynamics and variational inference under a single functional. This section reviews both, and reconstructs the second on the toy problems of Wild et al. to expose the machinery our own method inherits.

At the core of particle-based methods is an interacting physical system. To attempt to achieve the target, we apply forces that iteratively nudge our particles through the parameter space.

We can conceptualise the movement and final distribution of these particles as being governed by three primary forces:

1. The pull of the data (The Likelihood/Loss): A driving force that drags particles toward parameter values that minimise the empirical risk or self-information loss.
2. The anchor of the prior: A regularising force that pulls particles back toward our initial belief distribution $\pi(\theta)$, preventing overfitting.
3. The dispersion force (Repulsion/Noise): A mechanism that scatters the particles. This prevents the ensemble from collapsing into a single point estimate (MAP) and ensures the particles spread out to accurately map the uncertainty and variance of the target distribution. [7]

The diverse landscape of particle-based inference methods can be elegantly understood simply by examining how they implement, alter, or completely omit these three specific forces.

4.1 Stein varational gradient descent (SVGD)

Stein variational gradient descent (Liu and Wang, 2016) approximates a target $p(\theta)$ by deterministically transporting particles along the steepest descent of $\mathrm{KL}(q||p)$ within the unit ball of a Reproducing-kernel hilbert space. Explain the difference between L2 and RKHS and the norm perhaps

The three conceptual forces are directly seen in the SVGD update equation. Let our ensemble of M particles at iteration t be $\{\theta_j^{(t)}\}_{j=1}^M$. The particles are updated deterministically via a step size ε giving an update step

$$\theta_j^{(t+1)} \leftarrow \theta_j^{(t)} + \varepsilon \phi^\star(\theta_j^{(t)}). \tag{4}$$

The optimal perturbation direction $\phi^\star(\theta_j)$ is restricted to a Reproducing Kernel Hilbert Space (RKHS) governed by a positive-definite kernel $\kappa(\cdot,\cdot)$. It is computed as the empirical average over

[7] Forces 2 and 3 are not completely independent as we will see in Deep Ensembles

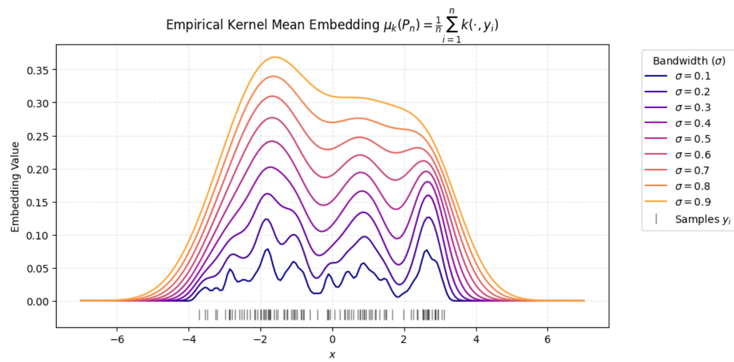


Figure 1: An Empirical Kernel Mean Embedding mapping discrete particles into a Reproducing Kernel Hilbert Space (RKHS). The bandwidth (σ) visually dictates the strictness of the RKHS smoothness penalty, illustrating how isolated data points are mathematically transformed into the continuous, interacting field that drives SVGD's repulsive force.

the entire particle ensemble:

$$\phi^\star(\theta_j) = \frac{1}{M}\sum_{k=1}^{M}\Bigg[\underbrace{-w\sum_{i=1}^{n}\nabla_{\theta_k}\ell(\theta_k,x_i)\kappa(\theta_k,\theta_j)}_{\text{1. The Pull of the Data}} + \underbrace{\nabla_{\theta_k}\log\pi(\theta_k)\kappa(\theta_k,\theta_j)}_{\text{2. The Anchor of the Prior}} + \underbrace{\nabla_{\theta_k}\kappa(\theta_k,\theta_j)}_{\text{3. Kernel Repulsion}}\Bigg]. \tag{5}$$

The first two terms represent the smoothed gradient (the score) of the generalised posterior $\nabla\log\pi_w$, acting as the driving force that drags particles towards high-density regions. In total, counting $\kappa(\theta_l,\theta)$ as the kernel-weightd average of the score (Also shown in Figure 1), this total attraction can be seen in one term $\kappa(\theta_l,\theta)\nabla_{\theta_l}\log p(\theta_l)$. The third term is the gradient of the kernel itself; which pushes neighboring particles apart and prevents mode collapse that ordinary gradient descent/assent would produce. The explicit deterministic repulsion is the defining mechanism of SVGD, contrasting it with other particle methods that rely on stochastic noise for dispersion.

The cost of SVGD is the pairwise kernel evaluation, $O(M^2)$ per iteration, and, in common with kernel methods generally performance degrades in higher dimensions as the kernel's ability to discriminate between particles weakens - Understanding the Variance Collapse of SVGD in High Dimensions - 'the kernel matrix essentially becomes uniform'

'In this work we attempt to explain the variance collapse in SVGD. On the qualitative side, we compare the SVGD update with gradient descent on the maximum mean discrepancy (MMD) objective;' - very important for my work. From Understanding the Variance Collapse of SVGD in High Dimensions

4.2 Deep Ensembles: Omitting the Dispersion Force

Wild et al. (2023) (and Lakshminarayanan et al., 2017??? for specifically deep ensembles not as a bayesian thing) establish a rigorous link between Bayesian inference, variational methods, and deep ensembles. They achieve this by recasting the non-convex training objective of deep learning as a convex optimisation problem over the space of probability measures $\mathscr{P}(\Theta)$. This aligns perfectly with the optimisation-centric perspective introduced in Section 1, and they analyse this space using Wasserstein gradient flows. Concretely, they minimise the free energy:

$$L(q) = \int_\Theta V(\theta)dq(\theta) + \frac{\lambda_1}{2}\iint_{\Theta^2}\kappa(\theta,\theta')dq(\theta)dq(\theta') + \lambda_2\int_\Theta q(\theta)\log q(\theta)d\theta, \qquad \text{where } V(\theta) = \ell_n(\theta) - \lambda_2\log\pi(\theta), \tag{6}$$

This objective can be broken down into three separate key ideas in a thermodynamic sum, similar to before. The first term is the external potential energy averaged over the particles where V is the potential. The potential energy $V = \ell_n - \lambda_2 \log \pi$ pulls the particles towards the data given by the loss function ℓ as well as the prior form $\lambda_2 \log \pi$. The second term, kernel interaction weighted by λ_1, is similar to the SVGD interaction term. Finally, the third term is a negative-entropy term weighted by λ_2.

Taking just the potential energy term and third term representing entropy, equivalent to setting λ_1 to 0, recoveres our π_w from before. Starting with our total free energy equation,

$$\begin{aligned}
L(q)\big|_{\lambda_1=0} &= \int_\Theta V(\theta)q(\theta)d\theta + \lambda_2 \int_\Theta q(\theta)\log q(\theta)d\theta \\
&= \int_\Theta \big(\ell_n(\theta) - \lambda_2 \log \pi(\theta)\big)q(\theta)d\theta + \lambda_2 \int_\Theta q(\theta)\log q(\theta)d\theta \\
&= \int_\Theta \ell_n(\theta)q(\theta)d\theta + \lambda_2 \int_\Theta q(\theta)\big(\log q(\theta) - \log \pi(\theta)\big)d\theta \\
&= \mathbb{E}_q[\ell_n(\theta)] + \lambda_2 \mathrm{KL}(q\|\pi).
\end{aligned}$$

Dividing this objective by λ_2 exactly recovers the Rule of Three objective from Section 1, where the generalisation learning rate is $w = 1/\lambda_2$. Consequently, the minimiser of this free energy is the Gibbs posterior $q^* = \pi_w \propto \pi(\theta)\exp\{-\ell_n(\theta)/\lambda_2\}$. As established previously, in the specific case where $\lambda_2 = 1$ ($w = 1$) and the loss is the self-information $-\log p(x_i \mid \theta)$, this objective recovers the standard Bayesian posterior π_B.

Furthermore, we can that when we set $\lambda_1 = \lambda_2 = 0$ we are left with only the $V(\theta)$ giving standard gradient decent and no dispersion of particles; the particles collapse onto the global minimiser of the loss. this is a deep ensemble.

Discretisation and Update Rule

As SVGD did and inline with particle approximations, to implement this infinite-dimensional gradient descent in practice, Wild et al. (2023) approximate the continuous-time Wasserstein gradient flow (Show optimal trasnport and what the wasserstein gradient flow is)using an ensemble of N_E interacting particles. The evolution of these particles in continuous time is governed by a stochastic differential equation (SDE) (of McKean-Vlasov type):

$$d\theta_n(t) = -\left(\nabla V(\theta_n(t)) + \frac{\lambda_1}{N_E}\sum_{j=1}^{N_E} \nabla_1 \kappa(\theta_n(t), \theta_j(t))\right)dt + \sqrt{2\lambda_2}\, dB_n(t), \tag{7}$$

for $n = 1, \ldots, N_E$, where $\{B_n(t)\}_{t>0}$ are independent standard Brownian motions, and $\nabla_1 \kappa$ denotes the gradient of the interaction kernel with respect to its first argument.

To simulate these trajectories computationally over a finite time horizon, the Euler-Maruyama discretisation (Appendix B in wild et al) method is applied. Given a discrete step size (or learning rate) $\eta > 0$, the update rule for the n-th particle at iteration t becomes

$$\theta_{n,t+1} = \theta_{n,t} - \eta\left(\nabla V(\theta_{n,t}) + \frac{\lambda_1}{N_E}\sum_{j=1}^{N_E} \nabla_1 \kappa(\theta_{n,t}, \theta_{j,t})\right) + \sqrt{2\eta\lambda_2}\, Z_{n,t}. \tag{8}$$

where $Z_{n,t} \sim \mathcal{N}(0, I)$[8] is standard Gaussian noise injected at each update step. Apparently. (still don't fully get this)

[8]The presence of η under the square root arises directly from the scaling properties of Brownian motion. When discretising the SDE, the infinitesimal increment $dB_n(t)$ is replaced by a finite time increment $\Delta B_n = B_n(t+\eta) - B_n(t)$. By definition, the increments of standard Brownian motion over a time step η are normally distributed with variance η, such that $\Delta B_n \sim \mathcal{N}(0, \eta I)$. To simulate this computationally using standard Gaussian noise $Z_{n,t} \sim \mathcal{N}(0, I)$, we must scale $Z_{n,t}$ by the standard deviation $\sqrt{\eta}$, yielding $\Delta B_n = \sqrt{\eta} Z_{n,t}$. Substituting this into the noise term of the SDE yields $\sqrt{2\lambda_2}(\sqrt{\eta} Z_{n,t}) = \sqrt{2\eta\lambda_2} Z_{n,t}$.

This update rule translates the thermodynamic properties of the free energy objective directly into algorithmic operations, recovering three distinct ensemble methodologies depending on the regularisation parameters:

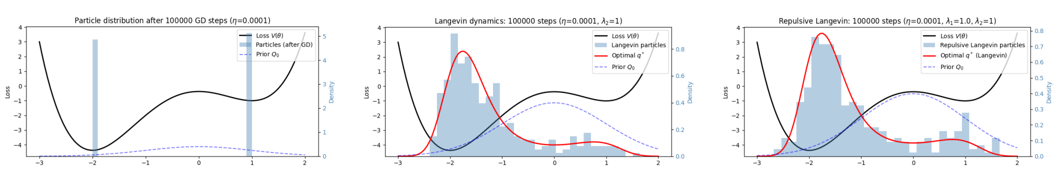
- **Deep Ensembles (DE):** As established above, setting $\lambda_1 = \lambda_2 = 0$ removes both the repulsive interactions and the injected noise. The particles evolve independently via standard gradient descent from their random initialisations, gravitating towards the local minima of the loss landscape.
- **Deep Langevin Ensembles (DLE):** Setting $\lambda_1 = 0$ and $\lambda_2 > 0$ yields independent Langevin diffusions. The particles do not interact, but the injected noise ensures they sample from the target Gibbs posterior rather than simply collapsing onto a mode.
- **Deep Repulsive Langevin Ensembles (DRLE):** When both regularisation parameters are strictly positive ($\lambda_1 > 0, \lambda_2 > 0$), the full update rule is engaged. Particles actively repel each other via the kernel κ while exploring the landscape with injected noise, guaranteeing convergence to the global minimiser of the free energy over the space of probability measures.

Toy Example: The Global Minimiser

To explicitly demonstrate the theoretical convergence guarantees of these interacting particle systems, Wild et al. (2023) construct a one-dimensional loss landscape featuring a deep global minimum alongside a shallower local minimum. By initialising particles from a standard Gaussian prior, they show that unregularised Deep Ensembles merely partition particles into these distinct modes based on their initial regions of attraction. In contrast, the noise and repulsive forces in fully regularised methods allow the ensemble to escape local traps and successfully represent the true global minimiser. The underlying non-convex loss function driving this behaviour is defined as an asymmetric polynomial:

$$\ell(\theta) = \frac{3}{2}\left(\frac{1}{4}\theta^4 + \frac{1}{3}\theta^3 - \theta^2\right) - \frac{3}{8}, \qquad \text{with prior } \pi(\theta) = \mathcal{N}(0,1).$$

To visualise the distinct behaviour of each ensemble method within this asymmetric polynomial landscape, we simulate the particle trajectories over 100,000 steps. Figure 2 illustrates the progression from deterministic collapse to fully regularised exploration.



(a) DE ($\lambda_1 = 0, \lambda_2 = 0$) (b) DLE ($\lambda_1 = 0, \lambda_2 = 1$) (c) DRLE ($\lambda_1 = 1.0, \lambda_2 = 1$)

Figure 2: Particle distributions after 100,000 steps across three regularisation regimes. **(a)** Driven solely by gradient descent, unregularised particles collapse entirely into point masses at the local and global minima. **(b)** The injection of noise allows particles to escape strict collapse and sample from the optimal Gibbs posterior. **(c)** Adding a repulsive kernel forces particles away from one another, resulting in a broader, flatter distribution that ensures more diverse coverage of the parameter space.

While the Deep Repulsive Langevin Ensembles capture a much broader representation of the space, it is crucial to understand that increasing this repulsive force introduces a fundamental trade-off. Figure 3 provides a comprehensive analysis of how scaling λ_1 shifts the algorithmic behaviour away from the true, unregularised posterior.

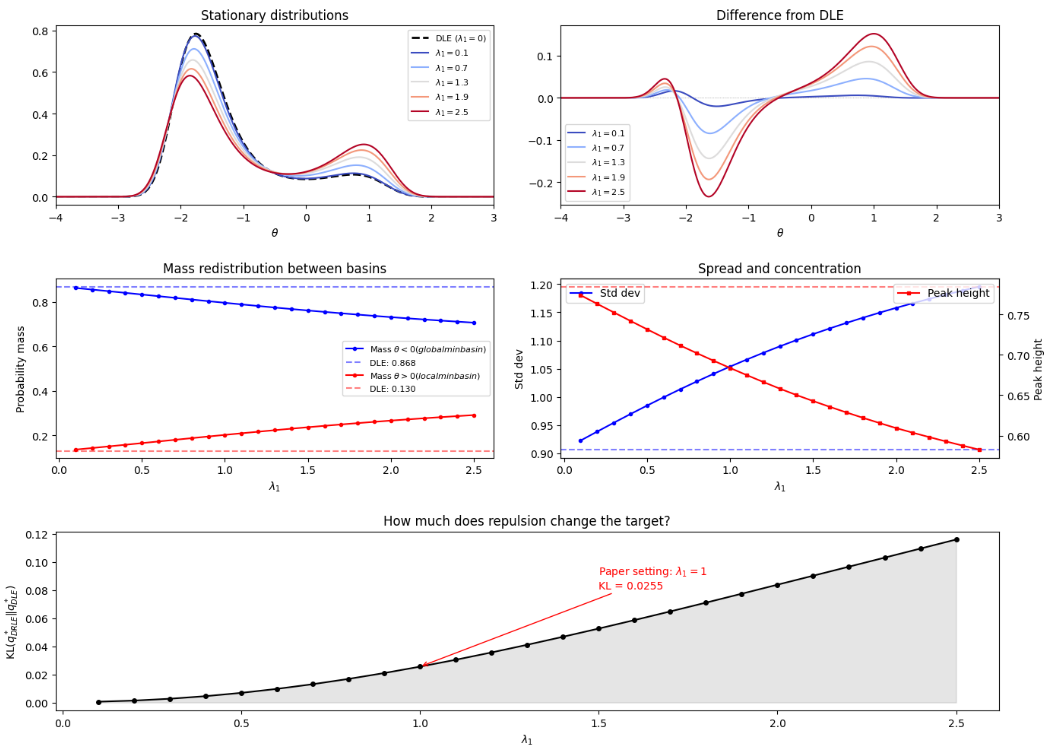


Figure 3: Quantitative impact of the repulsive regularisation parameter λ_1. As the repulsive force increases, probability mass systematically redistributes between basins, peak heights decrease, and the overall standard deviation widens. This increased particle diversity comes at the explicit cost of an expanding Kullback-Leibler (KL) divergence from the standard Deep Langevin Ensemble target.